

Practical 4

Aim: Use Python libraries like Pandas and Numpy to identify outliers using statistical methods (IQR, Z-score) and handle them by removing or capping.

Source-Code:

```
import pandas as pd
import numpy as np

data = {'Marks': [10, 12, 14, 15, 18, 20, 22, 25, 100]}
df = pd.DataFrame(data)
print("Original Data:")
print(df)

Q1 = df['Marks'].quantile(0.25)
Q3 = df['Marks'].quantile(0.75)
IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR
print("\nLower Limit:", lower_limit)
print("Upper Limit:", upper_limit)
outliers = df[(df['Marks'] < lower_limit) |
               (df['Marks'] > upper_limit)]
print("\nOutliers:")
print(outliers)
print("\nCleaned Data:")
clean_data = df[(df['Marks'] >= lower_limit) &
                 (df['Marks'] <= upper_limit)]
print(clean_data)
```

Output:

Original Data:

	Marks
0	10
1	12
2	14
3	15
4	18
5	20
6	22
7	25
8	100

Lower Limit: 2.0

Upper Limit: 34.0

Outliers:

	Marks
8	100

Cleaned Data:

	Marks
0	10
1	12
2	14
3	15
4	18
5	20
6	22
7	25